COMPUTER ENGINEERING

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	July 10, 2026	**DATE SUBMITTED:**	July 15, 2026
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ACTIVITY NO. 2 Basing C++ Programming

1. PROCEDURE OUTPUT

ILO A

```cpp
#include <iostream>

// Prototyping :
// 1. Create a function that will take two numbers and display the sum.
int sumOfTwoNum(int fNum, int sNum);
// 2. Create a function that will return whether variable A is greater than variable B.
int isAGreaterThanB(int aNum, int bNum);
// 3. Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.
bool logicalOperation(bool fVal, bool sVal);

int main() {
    // std::cout << "Hello World" << std::endl;

    std::cout << "The sum of the two numbers is : " << sumOfTwoNum(6, 7) << std::endl;
    std ::cout << "Is A Greater than B? : " << isAGreaterThanB(6, 7) << std::endl;
    std::cout << logicalOperation(true, false);

    return 0;
}

// Definition :
int sumOfTwoNum(int fNum, int sNum) {
    return fNum + sNum;
}

int isAGreaterThanB(int aNum, int bNum) {
    if (aNum >= bNum) {
        std::cout << "Yes! A is Greater than B.";
    } else {
        std::cout << "No!. B is NOT Greater than A.";
    };

    return 0;
};

bool logicalOperation(bool fVal, bool sVal) {
    std::cout << "\nFirst Value: " << fVal;
    std::cout << "\nFirst Value: " << fVal;
    // AND(&&), OR(||), NOT(!), XOR(^)
    std::cout << "\nAND(&&): " << (fVal && sVal);
    std::cout << "\nOR(||): " << (fVal || sVal);
    std::cout << "\nNOT(!) First Value: " << (!fVal);
    std::cout << "\nNOT(!) Second Value: " << (!sVal);
    std::cout << "\nXOR(^): " << (fVal ^ sVal);

    return true;
};
```

```
The sum of the two numbers is : 13
Is A Greater than B? : No!. B is NOT Greater than A.0

First Value: 1
First Value: 1
AND(&&): 0
OR(||): 1
NOT(!) First Value: 0
NOT(!) Second Value: 1
XOR(^): 11
```

ILO B

```cpp
#include <iostream>

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        // Constructor
        Triangle(double A, double B, double C);

    // *Function inside a class is called METHODS*
    void setAngle(double A, double B, double C);

    // Validator
    const bool validateTriangle();
};

int main() {
    // Driver Code
    Triangle set1(100, 30, 50);
    if(set1.validateTriangle()){
        std::cout << "The triange is Valid.\n";
    }   else {
        std::cout << "The triangle is NOT valid.\n";
    }
}

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

```
The triange is Valid.
```

ILO C

```cpp
#include <iostream>

void swapNumbers(int &fNum, int &sNum);

int main() {

    int fNum = 67, sNum = 76;

    std::cout << "Before Swapping: First Number = " << fNum << ", Second Number = " << sNum;

    swapNumbers(fNum, sNum); // Swapping

    std::cout << "\nAfter Swapping: First Number = " << fNum << ", Second Number = " << sNum;

    return 0;
}

void swapNumbers(int &fNum, int &sNum) {
    int temp = fNum;
    fNum = sNum;
    sNum = temp;
}
```

```
Before Swapping: First Number = 67, Second Number = 76
After Swapping: First Number = 76, Second Number = 67
```

```cpp
#include <iostream>

float kelvinToFahrenheit(float kelvin);

int main() {
    float temp;

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> temp;
    std::cout << "Temperature in Fahrenheit: " << kelvinToFahrenheit(temp)<< "F" << std::endl;
    return 0;
}

float kelvinToFahrenheit(float kelvin) {
    return (kelvin - 273.15) * 9 / 5 + 32;
}
```

```
Enter temperature in Kelvin: 2
Temperature in Fahrenheit: -456.07F
```

```cpp
#include <iostream>
#include <cmath>

float distanceBetweenTwoPoints (float x1, float y1, float x2, float y2);

int main() {
    float x1, y1, x2, y2, distance;

    std::cout << "Enter x1: ";
    std::cin >> x1;

    std::cout << "Enter y1: ";
    std::cin >> y1;

    std::cout << "Enter x2: ";
    std::cin >> x2;

    std::cout << "Enter y2: ";
    std::cin >> y2;

    std::cout << "The distance between two points is: " << distance << std::endl;
}

float distanceBetweenTwoPoints (float x1, float y1, float x2, float y2) {
    return sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
}
```

```
Enter x1: 4
Enter y1: 5
Enter x2: 6
Enter y2: 7
The distance between two points is: 0
```

```cpp
#include <iostream>
#include <cmath>

class Triangle{
    private:
        double sideA, sideB, sideC;
        double totalAngle, angleA, angleB, angleC;
    public:
        // Constructor
        Triangle(double A, double B, double C, double a, double b, double c);

    // *Function inside a class is called METHODS*
    void setAngle(double A, double B, double C);

    // Validator
    const bool validateTriangle();

    double computeArea();
    double computePerimeter();
    void triangleType();

};

int main() {
    // Driver Code
    Triangle set1(100, 30, 50, 3, 4, 5); // Angles and Sides
    if(set1.validateTriangle()){
        std::cout << "The triange is Valid.\n";
        std::cout << "Area: " << set1.computeArea() << std::endl;
        std::cout << "Perimeter: " << set1.computePerimeter() << std::endl;
    }   else {
        std::cout << "The triangle is NOT valid.\n";
    }
}

Triangle::Triangle(double A, double B, double C, double a, double b, double c){
    angleA = A;
    angleB = B;
    angleC = C;

    sideA = a;
    sideB = b;
    sideC = c;
}

void Triangle::setAngle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}

double Triangle::computeArea(){
    double s = computePerimeter()/2;
    return sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

double Triangle::computePerimeter(){
    return totalAngle;
}

void Triangle::triangleType(){
    if (angleA < 90 && angleB < 90 && angleC < 90)
        std::cout << "triangleType: Acute angle";
    else if (angleA < 90 || angleB > 90 || angleC > 90)
        std::cout << "triangleType: Obtuse angle";
        else  std::cout << "triangleType: Others(Right Traingle)";

}
```

```
The triange is Valid.
Area: nan
Perimeter: 6.95246e-310
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude Code**

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Double checking of formulas and helping for error**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was used only for double checking errors and formulas.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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